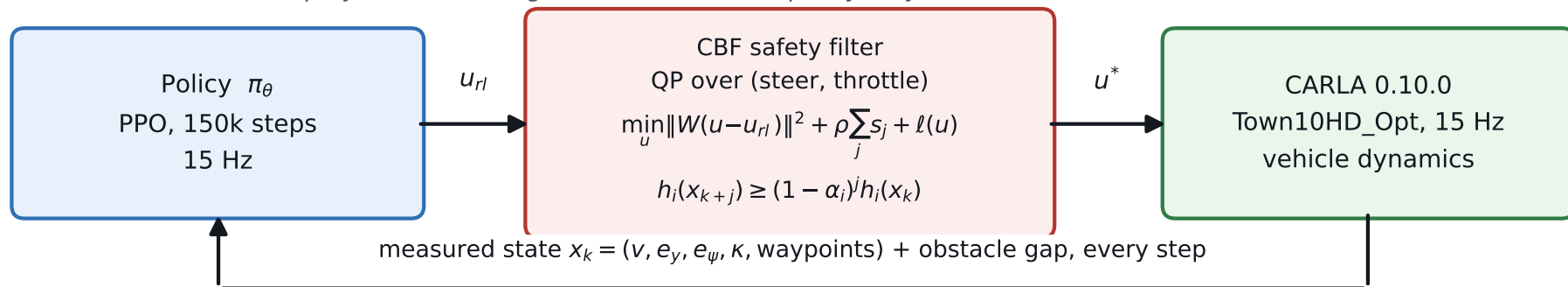
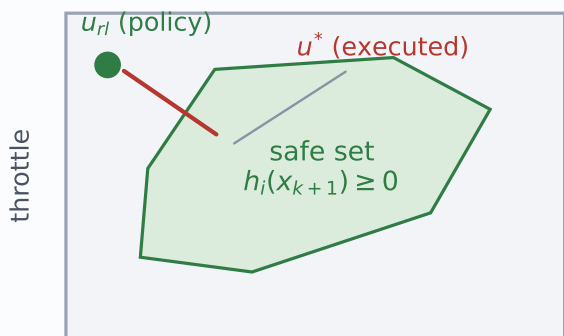


CBF safety filter for a learned driving policy

the filter is a projection: it changes the action of the policy only when that action would leave the safe set



what the filter does in one step



if u_{rl} is already inside the safe set, $u^* = u_{rl}$
(measured: $\|u^* - u_{rl}\|_\infty < 10^{-6}$)

the five barriers of this design

- lane, right $h_1 = (w/2 - m) - e_y$
- lane, left $h_2 = (w/2 - m) + e_y$
- speed $h_3 = v_{\max} - v$
- curve $h_4 = \sqrt{a_{\text{lat}}/\kappa} - v$
- headway to obstacle $h_5 = \text{gap} - (d_{\min} + \tau v)$

$w/2 = 1.75 \text{ m}$, $m = 0.30 \text{ m}$, $v_{\max} = 13.9 \text{ m/s}$,
 $a_{\text{lat}} = 2.5 \text{ m/s}^2$, $d_{\min} = 5 \text{ m}$, $\tau = 1.2 \text{ s}$

same trained checkpoint, filter off vs on:

worst $\min_i h_i$: $-0.53 \rightarrow -0.15$ worst $|e_y|$: $1.98 \text{ m} \rightarrow 1.60 \text{ m}$ (half-lane 1.75 m)